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Nanotechnology and Computer Science: Trends and advances

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ABSTRACT

Nanotechnology is the aptitude to perceive, measure, operate, and build materials at the nanometer scale, the size of atoms and molecules. Nanotechnology, is involved in many scientific and practical applications, including health, agriculture, electronic devices, computer science and many other fields. At the same time, computers play an essential role in simulating and analyzing the features of nanoparticles. The shrinking of electronic devices in due course will come to end, when the laws of physics avoid us from further diminishing the size of circuits. At any given time, the researchers will not be able to carry on with the current silicon processors, binaries and sequential. One of the most practical ways is to construct computers operating with radical difference: the quantum mechanics. The appearance of quantum computers will include a radical change in the hardware and software that we now know but will permit one to perform calculations that are unconceivable nowadays.

Memory is employed as one of the main fragments in diverse circuits design, therefore the design and optimization of random-access memory (RAM) cells have become one of the greatest remarkable investigation regions in Quantum-dot Cellular Automata (QCA) equipment. In this study, the researchers attempted to provide a systematic review of recent trends in the interaction between computer science and nanotechnology by highlighting two parallel axes, the first axis includes the use of nanotechnology in enhancing computer systems and devices, and the other one includes the role of computer science in promoting nanotechnology. Since the silicon technology is approaching its maximum capacity, which makes it essential to search for novel mechanisms that permit the production of nano-computers. Thus, this study confirms that computational nanotechnology is emerging as an essential engineering analysis tool for new device designs and modern applications.

1. Introduction

Nano is a prefix that indicates one billionth of a specified parameter. The prefix is often used in conjunction with the length scale to represent an exceedingly tiny amount of the length dimension [1]. The nanometer is a billion of a meter, for instance, the width of a piece of paper is about 100,000 nanometers thick, and the thickness of DNA is 2 nm, and the thickness of Nucleus is less than one nanometer [2]. For better visualization, the size of tennis ball in relation to earth as a nanoparticle is to a tennis ball (Fig. 1) [3]. In computers, IBM at 5 nm size, could squeeze 30 billion transistors into a chip has the dimension of a fingernail [4].

Nature and science have always included nanoscale particles. For example, viruses are naturally occurring nanoparticles from human activity, and as early as the 1920s, tires producers used carbon black to increase the performance of tires without being aware of nanoparticles [5,6].

Nanoparticles and nanostructured materials exhibit markedly different characteristics than their visible-scale counterparts [7]. For example, gold nanoparticles seem red rather than shiny yellow, while aluminum that is stable at the macroscale turns flammable at the nanoscale [8]. The distinct and unforeseen differences in attributes between the nanoscale and macroscale are owing to the dominance of quantum confinement and high surface area phenomena at the nanoscale. As material dimension shrinkages to the nanoscale, the fraction of atoms on the surface grows considerably [9]. This results in an escalation in the surface area to volume ratio. The augmented

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Fig. 1. Relation between tennis ball size to earth is same as nanoparticle to a tennis ball.

surface area to volume ratio, among other things, modifies the materials' reaction, mechanical, thermal, and catalytic properties [10]. Many researchers believe that Richard Feynman's archival talk, "There's plenty of Room at the bottom" 63 years ago, is considered the starting point and scientific glimpse of nanotechnology [11].

Since then, Nanotechnology has been a part of much of researches. It has made significant advances in agriculture [12], water purification [13], healthcare [14], transportation [15], energy [16], electronics [17], and environmental bioremediation [18]. To current time, preclinical studies have shown that nanotechnology-based tools are efficient against a number of diseases, including respiratory viruses [19], herpes virus [20], human papillomavirus [21], and HIV [22]. Due to the unique features of nanoparticles, which are particles their dimension ranging from 1 to 100 nm, they have generated considerable attention in the treatment of COVID-19 illness [23]. As a result, they may be considered as potential options for COVID-19 therapy [24]. Thus, nanotechnology offers several opportunities for enhancing the level of life in a sustainable way.

In terms of computer systems, nanotechnology has a promising future for developing computer components and functionality, at the same time, computers can simulate and enhance the process of nanotechnology development [25,26].

Previous research studies mentioned either the importance of nanotechnology in the progress of computer systems, or the role of computer applications in nanotechnology [27,28]. In this investigation, two aspects, regarding this interchangeable relationship between nanotechnology and computer science, were studied and analyzed systematically. The first aspect is the role of nanotechnology in developing computer systems as building memories, chips and sensors. While the second aspect is related to the role of computer systems applications and algorithms in simulating nanoparticles behavior and studying their properties. The ancient electronic memory schemes have mainly depended on the structure of transistors. Nevertheless, investigation into crossbar switch-built electronics have presented a substitute using reconfigurable interconnections between vertical and horizontal cabling arrangements to produce ultra-high capacity memories. Two frontrunners in this area are Nantero which has established a carbon nanotube based crossbar memory called Nano-RAM and Hewlett-Packard which has suggested the usage of mersister material as a forthcoming alternative of flash memories [29]. Investigators are emerging a type of memory chip with a predictable density of one terabyte of memory per square inch or superior. Decreasing the size of transistors utilized in integrated circuits. Scientists believed that it might be likely to put the power of the present computers in the palm of our hand [30].

The nature of the impact of nanotechnology on computer science and vice versa is illustrated in Fig. 2.

The following sections highlighting these interchangeable relations and roles. The role of nanotechnology in computer systems and the nanotechnology-based computer applications are the most important parts of this investigation. According to our best knowledge, this study serves researchers in both fields alike and opens new horizons for useful research that serves both fields and facilitates the process of understanding the nano world in a distinctive scientific way.

2. Role of nanotechnology in computer science

The request for amplified info storage densities has pushed silicon equipment to its confines and ran to a concentration on research on innovative ingredients and device assemblies, for instance magneto-resistive random access memory [31] and carbon nanotube transistors, for ultra-far-reaching integrated memory [32]. Electromechanical devices are appropriate for memory requests owing to their outstanding "on–off" relations and fast swapping appearances, however they comprise bigger cells and more complex structure procedures than silicon-based system [33].

In this section, the means by which nanotechnology can serve computer systems are presented. This can be done through a dense study for each of transistor building, quantum computer building, memory building and internet.

2.1. Nanotechnology and transistor building

Moore's law, designed by Gordon Moor in 1965, states that the amount of transistors on a chip twofold every two years [34,35]. Moore's law was applied for long time as shown in Fig. 3. However, Moore's law will not be able to applied for the next years due to physical limits that silicon transistors will have reached their physical limit for optimization and the size is approximately freeze in 5 nm [36]. Within this size electron jump nearby and turn up into undesirable places. Inside this tiny space, it turns out to be difficult to organize the fine-structure of the silicon wafer which is crucial to control the electrical properties. An additional factor that gradually ending Moore's law is the rising costs, since building extremely tiny materials, billions at a time, necessitates novel and enormously pricy apparatus [37].

Nanomaterials have become an attractive alternative for manufacturing the next generation of transistors due to their superior electrical characteristics, responsiveness, capacity to interface with other systems, high stability (including thermal stability), and natural tiny size [38]. Generally, there are binary main attitudes for synthesizing nanomaterials, bottom-up and top-down. In bottom-up approach the nanomaterials can be synthesized through the chemical reactions among the atoms, or ions or molecules. While top-down approach comprises the mechanical procedures to crush or break the bulk materials into numerous parts to form nanomaterials [39]. In addition, the bottom-up nanofabrication techniques accessible today enable the creation and integration of a large number of these tiny transistors onto chips and circuit boards.

Nanomaterials have been utilized to make a variety of different kinds of transistors. The field-effect transistor (FET) transistor, for instance, is often encountered due to its adaptability and compatibility with devices that have initiated to include nanomaterials into their fundamental design [40]. Circuit boards are a prime example, and the size of nanomaterial-based transistors plays a crucial role in this regard. Transistors are significantly used on circuit boards and processing components inside computers to convert and strengthen electronic-signals. The further transistors per unit area on a processor chip, the quicker it runs, resulting in an improvement in total speed and processing capability [41]. The ability to fabricate nano-transistors

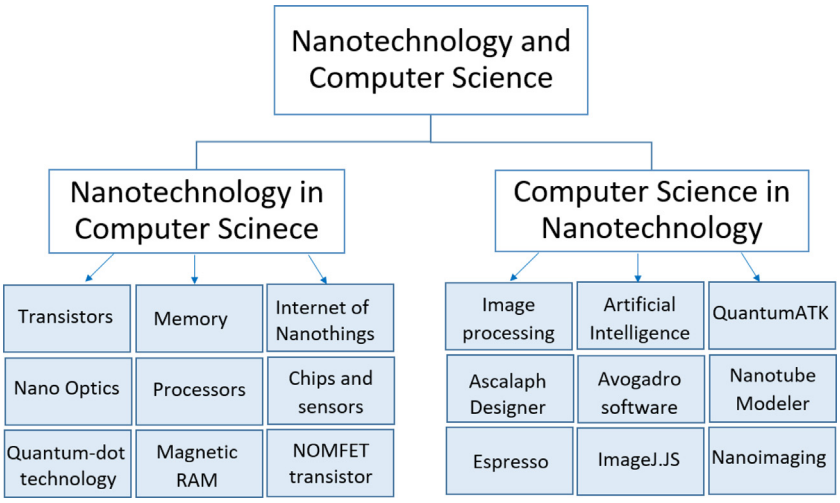

Fig. 2. Nanotechnology—Computer Science trends.

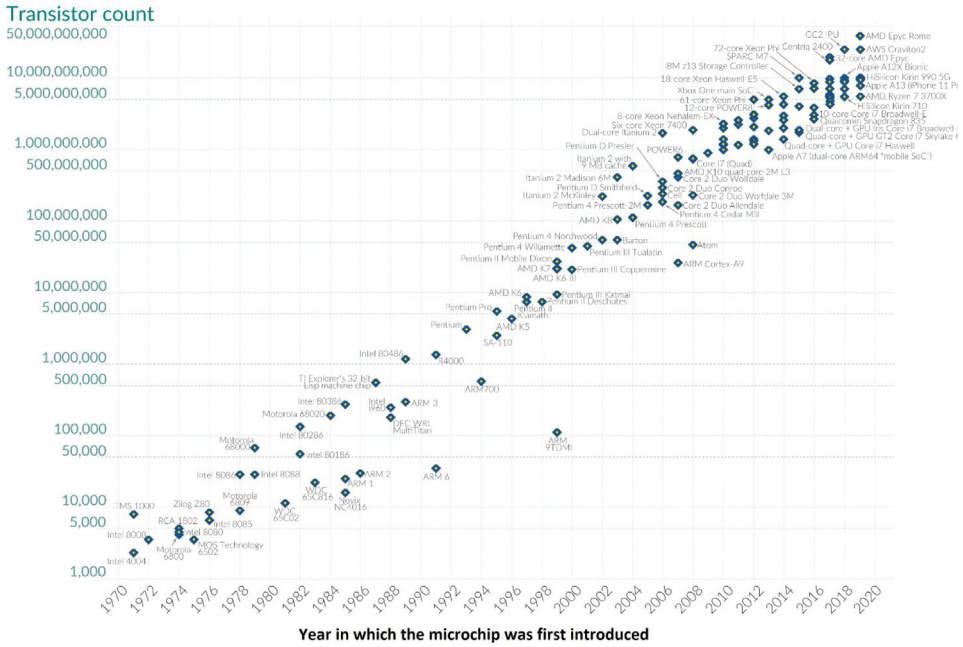

Fig. 3. The number of transistors per chip each two years.

allows integration of more transistors per space, thus enhancing chip performance, particularly with using highly conductive nanomaterials; for instance, carbon nanotubes (CNTs) [32]. Since carbon nanotubes can be synthesized atom by atom, they may be incorporated into chips in a customized manner that is consistent with the chip's standard design and the various components inside a device [42].

Utilizing carbon nanotubes is another technology that is assisting in the creation of more compact and quicker transistor designs, especially at IBM [43]. However, one of the primary drawbacks of carbon nanotube transistors is that, although their size is very tiny, their performance as transistors is inferior to that of silicon transistors [44]. However, researchers are making rapid progress in resolving this problem [45]. Carbon nanotube transistors capable of surpassing silicon transistors are being developed by scientists at the University of Wisconsin–Madison [46]. Indeed, their performance was found to be 1.9 times that of silicon transistors [47]. Carbon nanotube transistors, according to leading experts in the area, should be able to work at a pace five times faster and use five times less energy than silicon transistors [48]. That is, in an ideal case, however the true challenge in this field of computer nanotechnology is manufacturing and isolating carbon nanotube transistors [49].

Researchers from different sectors including, Taiwan Semiconductor Manufacturing Company (TSMC), the University of California, San Diego, and Stanford University, demonstrated a novel manufacturing procedure for carbon nanotube transistors that enables improved control [50]. This regulation is critical to ensure that transistors, in which operate as modifications in logical circuits, shut down completely on time. Carbon nanotube transistors have sparked renewed interest lately because to their potential to be minimized much more than silicon transistors and to enable the fabrication of stacked layers of circuitry much more simply than silicon allows.

Carbon nanotubes do not provide such a platform for deposition to begin [51]. They do not form oxide layers naturally, nevertheless, carbon dioxide (CO_2) and carbon monoxide (CO) are gases. Additionally, any faults in the nanotube that result in the formation of the required "dangling bonds" would impair its capacity to connect electricity [52]. However, much work has to be performed beforehand carbon nanotube devices can compete with silicon transistors. Numerous drawbacks

have been resolved independently but not yet merged into a single device [32].

On the hand single electron devices are talented candidates for progressive logic and memory circuits where the bits can be defined utilizing only a few electrons, leading to circuits with protection from statistical fluctuations in the number of electrons per bit and very low power consumption [53]. Nowadays single electron devices are thought to be among the top candidates to substitute standard complementary metal–oxide silicon field effect transistor technology at the end of the conventional semiconductor roadmap. Tunneling field effect transistor (TFET) [54], on the other hand, is a growing area of research nowadays since it is considered as a substitute for conventional complementary metal–oxide semiconductor (CMOS) by enabling in ultra-low power supply, and energy efficient computing. The transistor is based on band-to-band tunneling. The power enactment features of the TFET present not only the prospect of power reduction through low voltage operation but also present the challenge of meeting desires on die size and peak performance [55]. Obviously, chips only necessitating frequencies lower than the TFET–CMOS crossover frequency can use from a TFET implementation, since for these frequencies the TFET offers lower power at iso-performance or higher performance at iso-power [56].

2.2. Nanotechnology and quantum computers processor building:

IBM has made a significant advancement in semiconductor design and manufacturing with the announcement of the first chip in the world's using 2 nanometer (nm) nano-sheet technology [57]. Researchers lead by Volker Sorger, in George Washington University, has demonstrated a novel nano-photonic equivalent processor accomplished of resolving partial differential equations [58].

When electrons are stimulated, quantum dots emit just one wavelength of light. It is a novel material synthesized by the bottom-up process of nanofabrication [59]. They have the potential to revolutionize computer science and quantum information science by enabling the structure of a scalable, cost-effective, and fault-tolerant working machine. Quantum-dot technology has some promise application in this area [60].

In the upcoming years, quantum dots may be utilized as quantum bits and to form the foundation of quantum computers, in which the twofold rate of traditional computers is reproduced by quantum bits or qubits, which may be in any state-run between 0 and 1, concurrently both 0 and 1 [61]. Due to the quantum computer's ability to grip several circumstances concurrently, it is believed that it has the capacity to conduct a million calculations with one accord [62]. This would significantly increase the computer's speed. The growth of quantum computers until now in its infancy, meanwhile quantum computers that rely on quantum mechanical phenomena are at risk and faces several issues such as noise, coherence loss, and quantum bit loss [63].

2.3. Nanotechnology and memory building:

Utilizing hard drives in computers as memory, need more power and possess a higher failure rate than the solid-state memory, within stationary elements [64]. In theory there are two types of computer memory: primary and secondary. The term memory is used as a synonym for primary memory or as an abbreviation for a specific type of primary memory called random access memory (RAM) [65]. This type of memory is located on microchips that are physically close to a microprocessor. As a result, solid-state memory has gained popularity in smaller computers such as tablets [66]. This solid-state computer memory consumes less space, less battery power, and is less likely to be destroyed in the event of a device drop. It can be stated that, the density of solid-state computer memory is being increased by the use of nanotechnology [67]. Nanoscale transistors that are quicker, more influential, and progressively energy effective; momentarily our computer's whole memory might be stored on a sole miniature chip. Magnetic random access memory (MRAM) empowered by nanometer-scale magnetic tunnel-junctions that can rapidly and efficiently save even encoded data throughout a system power cut or smash, permit resume-play features, and collect vehicle coincidence data [28].

Quantum-dot Cellular Automata (QCA) is innovator nanotechnology that possesses increased attention in the investigation communal for suggesting little power, little area, and extremely effective numerical circuits in next generation computer policies. Memory is used as one of the main parts in dissimilar circuits scheme, therefore the policy and optimization of random-access memory (RAM) cells have developed one of the hottest topic research areas in QCA machinery. Mubarakali et al. [68] proposed a novel D-latch based nanoscale RAM cell, which is appropriate for operation the effective single layer QCA based paths. Moreover, through using this memory cell, one-by-four RAM can be obtained. These circuits are intended utilizing QCA fashionable and practical precision of replication consequences can be verified. As stated by Mubarakali et al. the familiarized nanoscale structure contains small dormancy and a tiny number of cells contrary to its preceding equivalents [68].

The novel requests and strategies that are presently under exploration by researchers and engineers, have ability to bring rising to each field they will be useful. In the field of computer science, novel quantum computers are intended. Motorized and atmosphere manufacturing have a request for "systems-on-a-chip", in which diminishment permits all electronic classifications, like computer, memory, management, steering, communication, power, sensors, actuators, to be appropriate on a small chip. Such arrangements cannot be appeared with existing equipment, but their arrangement will bring about a novel age of computer and aeronautics systems, space transference and investigation [69].

The researchers discovered that they may frequently switch each memory cell's status between conductive and nonconductive without affecting the material properties. They also believe that, they may obtain higher memory densities by stacking many layers of nanowire arrays together in the manner of a triple-decker club sandwich [70].

Another technique has been industrialized for increasing the density of memory devices is to store data on magnetic nanoparticles [71]. Investigators at North Carolina State University were created rising collections of magnetic nanoparticles, named nanodots, which are around 6 nm in length. Each dot would hold information based on its magnetization. Utilizing billions of these 6 nm length dots in a memory device has the potential to significantly boost memory density [71].

More or less of the nano-electronics regions under improvement for instance conjoining gold nanoparticles with organic molecules to produce a transistor recognized as a NOMFET (Nanoparticle Organic Memory Field-Effect Transistor). Also, by means of nanosized magnetic rings researchers can make Magneto-resistive Random Access Memory (MRAM) to produce a memory density of 400 GB per square inch. Similarly, through utilizing Bucky-balls researchers can build dense, low power memory devices.

2.3.1. Emerging memory devices

The novel memory technologies are required to maintain the data without static power supply, specifically the data is not volatile. The so-called non-volatile RAM technologies have been proposed and developed as the replacement of dynamic-RAM. For the meantime, the gap in the calculating speed needs to be filled as well. A search for novel memory technologies has been conducted for more than three decades. Numerous types of non-volatile RAM have been anticipated and thoroughly studied. The emerging non-volatile RAM candidates includes; Redox RAM (ReRAM), phase-change RAM (PCRAM), ferroelectric RAM (Fe RAM) and magnetic RAM (MRAM). To summarize, Table 1 lists the key parameters of each type of novel RAM technologies and compare them with the typical memories.

Memory system organization is a main element in determining computer enactment. Diverse memory technologies for instance dynamic-RAM, static-RAM, and hard drives, compromise trade-offs in size,

Table 1
Performance of emerging memories and their comparison with typical memories.

	Memory type	Cell size	Properties	Scalability	Ref.
Traditional memory devices	Static-RAM	64,000 nm^2	Read/write time < 2 ns	Good	[72]
	Dynamic-RAM	2900 nm^2	Read/write time 30–50 ns	Limited	[73]
	NAND Flash	784 nm^2	Read/write time 10^3–10^6 ns	Limited	[74]
Emerging memory devices	Ferroelectric-RAM	880 nm^2	Read/write time 5–10 ns	Limited	[75]
	Resistive-RAM	88 nm^2	Read/write 20–50 ns	Medium	[76]
	Phase change-RAM	127.5 nm^2	Read/write 2–100 ns	Limited	[77]
	spin-transfer torque-MRAM	1870 nm^2	Read/write 10–20 ns	Good	[78]
	Spin–orbit-torque-MRAM	3520 nm^2	Read/write < 10 ns	Good	[78]

Table 2
Comparison of memory technologies.

Memory type	Cell size	Advantages	Disadvantages	Feasibility	Ref.
Complementary metal–oxide semiconductor (CMOS) memory	9600 nm^2	Retains both functions, even when the computer is turned off	Some circuits are not practicable Difficult to implement	Yes	[79]
Dynamic-RAM	350 nm^2	Developed technology, High density	High energy consumption	Yes	[80]
Static-RAM	64,000 nm^2	Developed technology, Low energy consumption	Low density	Yes	[81]
Phase change-RAM	127.5 nm^2	Ultra-high density, High access speed	Experimental technology	Not-yet	[82]
NAND Flash single-level cell	784 nm^2	Developed technology, Low power consumption	Scalability concerns	Yes	[83]
NAND Flash multi-level cell	392 nm^2	Developed technology, Low-cost manufacturing	High power consumption, Low write and read speed	Yes	[84]
NOR Flash	1293 nm^2	Developed technology, High read speed, Low energy consumption (read)	High power consumption (write)	Yes	[85]
Racetrack	200 nm^2	Good scalability, High read/write speed	Experimental technology	Not-yet	[86]
Giant magneto-resistance (GMR)	500 nm^2	Excellent scalability	Experimental technology	Not-yet	[87]
2D Material	0.1 mm^2	Low power dissipation, they have ultimate limit of vertical downscaling	Engineering and processing issues	Yes	[88]
Ferroelectric-RAM	55 nm^2	Nondestructive readout, small cell size and scalability	Template techniques need a gentle way to remove the template to minimize damage	Yes	[89]

speed, and cost. Complementary Metal Oxide-Semiconductor (CMOS) as it is being utilized nowadays in state-of-the-art processors. Nevertheless, modern research has revealed that certain types of nanoscale electronics, when utilized in crossbreed configurations with current CMOS, could be beneficial in generating novel computing structures, in addition to high density memory, and hybrid and neuromorphic structures. Neuromorphic structures consist of electronic circuits and devices built using design principles that are based on those of biological nervous systems.

Table 2 summarize the memory technologies from CMOS to state-of-the-art emerging memories. The material advancements, shortcomings and solutions at each stage for each memory device also highlighted in Table 2.

2.4. Internet of nanothings

The first thought of internet of nanothings was suggested in 2010 by Akyildiz and Jornet in their article titled "The Internet of Nanothings" [90]. Internet of nanothings is the novel development that mergers internet of things through nanotechnology. The Internet of nanothings is relying upon artificial biology and nanotechnology implements that permit the engineering of biological entrenched computing procedures occasions. Nano sensors have become one of the key investigation arenas in nanoscience owing to possessing real explanations in many fields including computing systems [91].

Two architectural concepts were examined: intrabody nanonetworks for remote treatment and the future linked workplace [92]. The nanoscale, on the other hand, is the ordinary province of molecules, proteins, DNA, organelles, and the primary modules of cells [93]. Current biological nano sensors and nanoactuators, for example, offer a link among biological phenomena and electronic nanodevices, which may be leveraged over this novel networking pattern [94]. Also, every element initiates in an office, including its interior machineries, is equipped with a Nano transceiver that enables them to remain continually linked to the Internet, allowing for the tracking of all user possessions.

The Internet of Nano-Things' network architecture is composed of four components:

- Nano-nodes: The smallest nanomachines capable of performing basic computations.
- Nano-routers: Used to aggregate information from nano-nodes and control their behavior
- Nano-micro interface devices: Connects the information from nanoscale routers to conventional communication networks and conversely
- Gateway: permits the complete system to be remotely controlled through the Internet.

Theoretically, implementing transceivers to illustrate the viability of internet of nano-things has been shown by research on graphene radios

operating at terahertz frequencies [95]. Thus, intrusive monitoring in situ with implanted biosensors is possible. Additionally, they may be used to monitor the environment, such as waterways. Due to the large bandwidth of graphene-based transceivers, they have been shown to work at one terabit per second, but the auxiliary electrical components required to make the nano-transceiver a reality are still being explored. Interfacing internet of nano-things with current micro-devices is critical if it is to become ubiquitous. Thus, further research should be directed toward this goal, particularly in the industrial, biomedical, and industrial spheres. Significant hurdles in the disciplines of electromagnetic channel modeling at this biological cellular scale, as well as the essential supporting networking protocols, must be solved [95].

Because security is a major concern for the internet of nanothings, owing to the enormous amount of nano sensors through restricted compute aptitudes that preclude the use of advanced security algorithms. There is a need for seamless integration of hardware and software explanations into the current arrangement in order to handle the obstacles associated with its successful implementation, particularly in terms of security [96].

Although internet of nanothings is not yet at the level of other internet of things systems in terms of development, it is garnering attention from the communication and medical industries [97]. According to the modern investigation, internet of nanothings marketplace is estimated to raise from about 5 billion USD in 2016 to 10 billion USD by 2020. Also, at an expected rate of more than 24% for the recent estimate period of 2016 to 2025 [98]. Internet of nano-things arrangement can get installed by mixing of nano devices and several other technologies like internet of things, sensors network, cloud work out, and great data analytics and so on [99].

2.5. Optical nano technology

Nanomaterials have been utilized to generate a range of modules within fiber optic cables. These comprise optical filters and Bragg gratings, which are either utilized on their own within fiber optic cables or to construct other modules, for instance fiber optic anemometers. Numerous nanomaterials are utilized to construct diverse parts of fiber optic cables since they interact strongly with the propagating light [100]. Other areas where nanomaterials are utilized to create photonics-based devices include fiber optic rectennas to change electromagnetic waves into an electric current, more advanced magnetic recording devices, and ways of refining the light intensity in spectroscopy and solar cell technologies, to name a few. In recent years, a wide range of nanomaterials have been used due to their strong interaction with light and their ability to manipulate the properties of light. Thus, many more technologies are being developed that use high-tech nanomaterial optics and photonic components [101].

Nanotechnology has inspired new directions for fiber optics research, including fiber optic nano-photonics [102]. Once we consider about optical fibers, we often reflect of optical propagation via waveguides with diameters larger or equal to the wavelength of light [103]. For instance, an optical fiber with sufficient refractive index contrast may perform functions other than merely guiding light from one place to another [104]. Nanotechnology can be utilized in many areas of application [105]. Nano-features, in their most commonly known form, offer a different mechanism for light confinement in supposed photonic crystal fibers [106]. Despite continuous development in some areas, advancement in optical loss has stalled due to severe restrictions. Moreover, it appears that the low delay caused by air transport becomes relevant only when the fiber length approaches 1 km [107]. Due to constrained power budgets, this necessitates an increase in optical attenuation from existing values. As a consequence, the focus in communications has switched to the longstanding potential for resonating core fibers to give significant gains in communications capacity owing to their reduced operative nonlinearity [108].

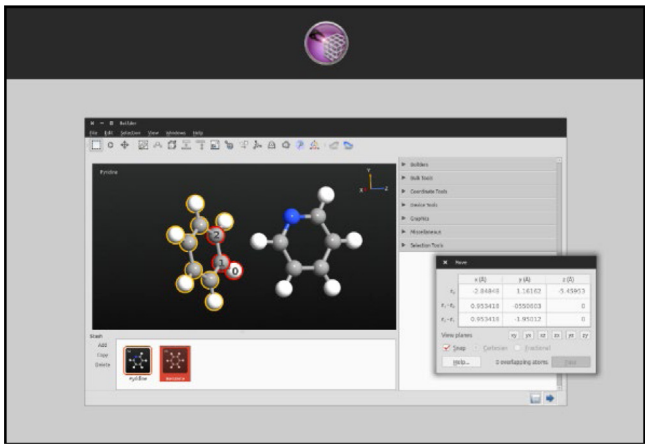

Fig. 4. QuantumATK Screen.

Another way to use nanotechnology in optics is to use it to create very small silicon devices, such as those made from luminous nanoparticles five nanometers in diameter or less [109]. Quantum confinement phenomena enable the gadget to produce light at that scale. However, electrical connections at that size are not now viable and may jeopardize the nanomaterial's optical activity, as well as provide very poor electrical conductivity [110].

2D-crystals are developing materials for future nano-electronics, the community of which has been looking for applicant channel materials that possess adequate electronic band gap, high carrier mobility, and good channel-lead contact. A few-layer black phosphorus possesses a direct band gap, which is tunable from 1.51 eV of monolayer to 0.59 eV of 5-layer. Whereas, the hole mobility, for instance, increase from 337 cm^2/V.s of monolayer to 2722 cm^2/V.s of 5-layer. Additionally, directionally discerning optical transitions were found that important light absorption occurs at energies in the infrared range along a definite direction, while there is no considerable absorption below 2.0 eV for the other two directions. These consequences make few-layer black phosphorus a talented nominee for upcoming electronics and optoelectronics devices [111].

3. Role of computer science in nanotechnology

This section highlights the contribution of computer science in nanotechnology in terms of computer simulation, Nano-image analysis enhancement and the usage of artificial intelligence in nanotechnology.

3.1. Computer applications:

Meanwhile investigational study at the nanoscale is very pricy and technologically challenging, researchers are urged to utilize computational approaches to simulate the behavior of nanostructures [112–115]. Models proposed that the usage of high-k dielectrics in the capacitors will rise the capacitance to the desirable stages for active random access memory requests. In the following section, different nanotechnology-related applications are listed.

3.1.1. QuantumATK

QuantumATK is an integrated set of atomic-scale modeling tools developed since 2003 by professional software engineers in collaboration with academic researchers. QuantumATK (Fig. 4) provides a simulation environment for developing unique materials with the optimal qualities for new or existing goods and systems. It is used in universities to explore the nanostructures of particles from the atomic level [116].

QuantumATK is written in the C++ and Python programming languages. Nearby 80% of the code is written in Python, with just

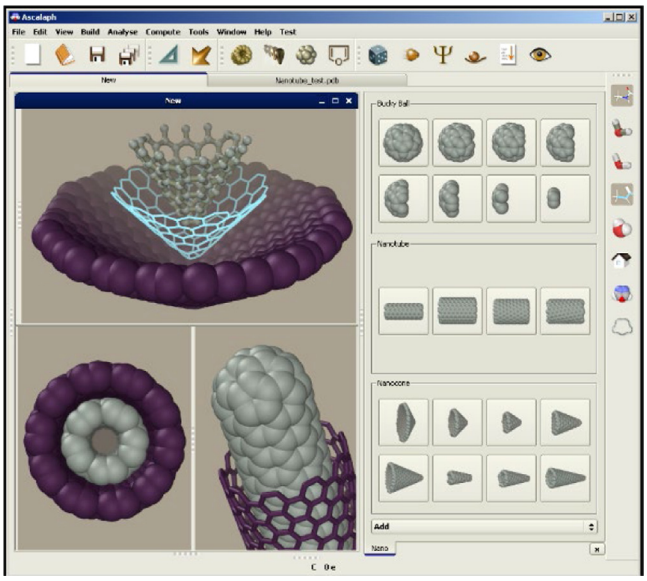

Fig. 5. Ascalaph Designer screen.

the lowest-level mathematically demanding sections written in C++. Python's usage of elevated physics and mathematics collections enables the rapid development of Quantum's comprehensive features. Python scripting enables the creation of sophisticated processes for automating complicated simulations and analyses [116]. Nevertheless, more or less simulations involve a significant amount of time-consuming computation activities that are then merged to produce a final result, making scripting such processes problematic. For example, if the computer fails in the midst of a script's loop, how do you resume the script at the correct moment in the script's loop? Or maybe some more activities are required after the conclusion of a bespoke simulation; how to mix the previously computed data with the new data? QuantumATK has proposed a framework termed a research object to facilitate such simulations. The study object manages sophisticated simulations that need the execution and combining of a number of fundamental activities. It enables concurrent execution of fundamental operations and resumption of the computation if it is halted before completion [116]. QuantumATK can be applied for atomic-scale modeling in semiconductor physics, catalysis, polymer materials, battery materials, and other fields.

3.1.2. Ascalaph Designer

Ascalaph Designer (Fig. 5) is a general-purpose tool for doing dynamic simulations of molecular systems [117]. Under a single graphical interface, prominent programs' approaches of classical and quantum mechanics are represented as their own implementation of molecular dynamics [118].

Ascalaph Designer is written in C++ and it is a piece of software that enables the creation and simulation of molecular representations and their transformation into a three-dimensional atmosphere. Additionally, it offers a graphical interface for modeling traditional and quantum molecules using Fire *y* and MDynaMix/MGE [119]. Together the graphical surroundings in which the simulant is constructed deliver devoted applications for the interface of a great amount of atoms nevertheless lack the computational power that these interpretation devices provide on bunch systems or devoted hardware, for example NVIDIA Corporation-Molecular Dynamics [120]. Ascalaph graphics is the basic graphical interface module providing for the main window design, three-dimensional graphics, information windows, and error message windows. It, also, provides interaction with the quantum mechanical program PC GAMESS. Ascalaph Designer provides the methods to build and edit molecular models, geometry optimization, molecular dynamics simulations.

3.1.3. Avogadro software

Avogadro (Fig. 6) is a permitted, open basis molecular editing and conception program for use in computational chemistry, molecular demonstrating, bioinformatics, materials science, and connected fields for Mac, Windows, and Linux [121]. Avogadro 1.1 introduces a novel nanotube builder that is relies upon the famous TubeGen code and website from the Doren group at University of Delaware [122]. There is a new choice for the nanotube builder in the Build menu [123]. The constraints regarding Avogadro it can be applied to fix or ignore a specific selection of atoms in a molecule, as well as to fix distances, and angles. The limitation of Avogadro is there is not Python bindings available for Windows, except when using the very out of date version 1.0.1, which can prevent one from using some of the more advanced features [124]. Avogadro offers a semantic chemical builder and platform for visualization and analysis. For users, it offers an easy-to-use builder, integrated support for downloading from common databases such as PubChem and the Protein Data Bank, extracting chemical data from a wide variety of formats, including computational chemistry output, and native, semantic support for the CML file format [121].

3.1.4. Nanotube modeler

Nanotube Modeler is an advanced software application designed specifically for helping you generate x, y, and z coordinates for nanotubes and nanocones, and analyze them using the integrated viewers or an external program of your choice. In addition, the tool is able to create buckyballs and graphene sheets using interactive graphs [125]. Nanotube Modeler gives the users possibilities of hiding or showing the atoms and bonds from the working environment, adjust the size of atoms and bonds, rotate the object around the *x*, *y* or *z* axis, and choose between several stereo modes (parallel, cross-eye or anaglyph) [126].

The Nanotube Modeler application (Fig. 7) generates xyz coordinates for Nanotubes and Nanocones [125]. The generated geometries may be seen directly in the integrated viewer or through contacting a third-party viewer software. This software is relying upon my JNanotubeApplet, nevertheless has enhanced and expanded functionality [127]. One of the software plugins is generating Nanotube Hetero-Junctions CoNTuban algorithm that is used for linking two arbitrary carbon nanotubes [128].

3.1.5. Espresso

Espresso is a popular android mobile testing framework designed for white box testing (Fig. 8). Google released Espresso in 2013. This open-source testing framework is now available as a part of the android support repository. The name 'Espresso' has Italian roots which means coffee. It is a simple and flexible testing framework [129]. It offers Application Programming Interfaces (APIs) that enable you to write User Interface (UI) tests. Espresso is coming from the IT giant Google. Espresso is highly popular among android testing tools users and easy for developers. Espresso supports JUnit4 and is backward compatible [130]. It supports out-of-app testing too. Tests are executed with speed. The Espresso Recorder is a handy tool to create UI tests without writing the code. It supports web component testing too [131].

To summarize, Table 3 lists the key parameters of each type of the above-mentioned software and key features, usage, and prominent of each of them.

3.2. Image enhancement with digital signal processing

Nanotechnology is one of the many uses for non-comprehensive image processing. Scanning Electron Microscope (SEM) and Atomic Force Microscope (AFM) are effective characterization tools for nanostructures. Different image processing techniques can be used for this purpose, as image enhancement tools, image segmentation, and computing statistical values [132,133].

Digital image processing was utilized to see the sights the structural characteristics of carbon material [134]. Also, different researches

Fig. 6. Avogadro Screen with Nanotube Builder.

Table 3
Computer program software utilized for nanoscience and nanotechnology.

Computer software	Initial release	Key features	Applications	Prominent points
QuantumATK	2003	written in the C++ and Python programming languages	Can be used to explore the nanostructures of particles from the atomic level	It enables concurrent execution of fundamental operations and resumption.
Ascalaph Designer	2015	written in C++	Can be used for Molecular model building: polymers, nanotubes, proteins, nucleic acids	It provides interaction with the quantum mechanical program
Avogadro software	2008	written in C++	Can be used to fix or ignore a specific selection of atoms in a molecule, as well as to fix distances, and angles	Avogadro offers a semantic chemical builder and platform for visualization and analysis
Nanotube Modeler	2005	written in C++	Creation of Nanotubes, Nanocones, Buckyball, Graphene Sheet	Nanotube Modeler gives the users possibilities of hiding or showing the atoms and bonds from the working environment
Espresso	2013	written in C++	It offers Application Programming Interfaces (APIs) that enable you to write User Interface (UI) tests	The Espresso Recorder is a handy tool to create UI tests without writing the code

Fig. 7. Nanotube Modeler.

investigated, using TEM analysis, to study the microstructure of carbon materials. Sun and Barnard described a modest method of adapting materials structures in to exclusive image based patterns appropriate for image processing approaches, that does not necessitate particular pre-assessment of the data and assortment of descriptors by the operator [135].

Additionally, a MATLAB software was built with the objective of assessing the atomic lattice features of pictures acquired using high resolution electron microscopy [136]. The anodic oxide generated on pure Al TLC (Thin-Layer Chromatography) film without pre-annealing is researched in 2016 [137]. Through digital image analysis, the impacts of anodizing duration, concentration, and temperature on the structural characteristics of the oxide films are investigated.

Mohammad [138] utilized the kmeans scheme to regulate the optimum edge by means of arithmetical estimation. This method is methodically investigated for the SEM image in silver nanostructure. He showed that, the obtained images through SEM analysis are presentable to be analyzed within this technique. The kmeans procedure categorizes the set of data assumed to k groups built on convinced extents of induced spaces. Kmeans tools is the most extensively utilized amongst all assembling procedures. Thus, it is one of the communal methods utilized in arithmetical data investigation, image study, neural networks, arrangement study and biometric statistics. We can also add